Practical - 11

Student Management CRUD with Search and Filter

Problem Definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MySQLi statements and clear success/failure messages.

A. Theory

1. What is CRUD?

Answer: CRUD stands for Create, Read, Update, and Delete. These operations cover the basic lifecycle of database records.

2. Why use MySQLi prepared statements?

Answer: Prepared statements help keep user-provided values separate from SQL syntax and reduce SQL injection risk.

3. How do search and filter differ?

Answer: Search matches a user-entered phrase across selected fields. A filter restricts results using a condition such as course or status.

4. Why confirm deletion?

Answer: Confirmation helps prevent accidental removal. The server must still validate the record ID and permissions.

B. Implementation / Procedure

1. Database setup

Create a students table with id, name, email, course, phone, and created_at. Add suitable constraints, including a unique email if appropriate.

2. Create and Read

Use validated POST input and prepared INSERT statements to add records. Use SELECT to display students in a table.

3. Update

Load the selected record by ID, populate an edit form, validate submitted changes, and execute a prepared UPDATE statement.

4. Delete

Confirm the action in the interface, then send a POST request. Validate the ID and execute a prepared DELETE statement only for authorized users.

5. Search and filter

Use prepared SELECT queries with LIKE conditions for name/email/course and an optional course filter. Escape displayed values with htmlspecialchars().

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C. Test Cases

Test Case	Input / Action	Expected Result
Add student	Submit valid student details	New record appears in the list
Edit student	Change course or phone	Updated values appear
Delete student	Confirm delete for a record	Record is removed and success message appears
Search	Search a known name/email	Only matching records are shown
Filter	Choose a course	Only students in that course are shown

D. Key Questions / Analysis / Interpretation

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

Answer: SELECT displays records, INSERT adds a student, UPDATE edits an existing record, and DELETE removes a record after confirmation. Each operation reports success or failure.

2. Is search by name/email/course implemented?

Answer: A search field can match name, email, or course using a prepared query. A course filter can narrow the results further.

3. Are delete operations confirmed before execution?

Answer: Yes. A confirmation prompt is shown before the delete request is submitted to prevent accidental deletion.

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Work (Screenshots)

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StudentHub Portal

Home | About | Students CRUD | Login | Dashboard | Events | Profile | Contact | FAQ | Feedback

Light / Dark

Practical 11: Student Management CRUD

Add, View, Edit, Soft Delete, Search, Filter & CSV Export

+ Add New Student Download CSV Export

Search name, email, or mobile... | All Courses | All Years | Filter

Total Active Students: 22

#	Name	Email	Mobile	Course	Year	Gender	Actions
1	archi bhatt	25dce009@charusat.edu.in	9316806552	B.Tech Computer Engineering	Second Year	Female	Edit | Delete
2	vatsal makwana	25dce050@charusat.edu.in	9726093919	B.Tech Computer Engineering	Second Year	Male	Edit | Delete
3	Kavya Barot	kavya.barot@charusat.edu.in	9825010203	B.Tech Information Technology	First Year	Female	Edit | Delete
4	Pranav Makwana	pranav.makwana@charusat.edu.in	9825001234	B.Tech Mechanical Engineering	Second Year	Male	Edit | Delete
5	Meera Solanki	meera.solanki@charusat.edu.in	9825090123	B.Tech Civil Engineering	First Year	Female	Edit | Delete
6	Siddharth Vyas	siddharth.vyas@charusat.edu.in	9825089012	B.Tech Computer Engineering	Fourth Year	Male	Edit | Delete
7	Tanvi Raval	tanvi.raval@charusat.edu.in	9825078901	B.Tech Information Technology	Second Year	Female	Edit | Delete
8	Dhruv Panchal	dhruv.panchal@charusat.edu.in	9825067890	B.Tech Computer Engineering	Third Year	Male	Edit | Delete
9	Kinjal Pandya	kinjal.pandya@charusat.edu.in	9825056789	B.Tech Mechanical Engineering	Third Year	Female	Edit | Delete
10	Manan Modi	manan.modi@charusat.edu.in	9825045678	B.Tech Civil Engineering	Fourth Year	Male	Edit | Delete
11	Isha Bhatt	isha.bhatt@charusat.edu.in	9825034567	B.Tech Information Technology	First Year	Female	Edit | Delete



N. GitHub Repository Link

https://github.com/vatsal-1950/StudentHUB.git